Airport Noise Model

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Course: ES115 – Engineering Computer Applications

Module 2: Physics & Chemistry Applications

4-27-26

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Problem Overview / System Description

- Airport noise is one of the most regulated environmental impacts in civil and aerospace engineering. Airports are required to demonstrate that surrounding communities stay below mandated noise thresholds. Engineers use noise models to design flight procedures, select quieter aircraft, and plan new runways.
- This system models a noise monitor exposed to aircraft, where the aircraft moves along a straight horizontal flight path at constant altitude and speed to find the noise level in decibels at that moment.
- This model reflects the calculation behind the FAA's Integrated Noise Model (INM), which airports use for environmental impact assessments.

Engineering Objective

- The model analyzes how the noise level at a fixed ground receiver changes over time as an aircraft flies overhead.
  - Predicts the instantaneous sound pressure level in decibels along a given plane's flight path
  - from 5 km before the closest point to 5 km after.
- **Driving Question:** Given an aircraft's sound power level, altitude, speed, and the atmospheric absorption of the air it is flying through, what is the peak noise level experienced on the ground and how does that level change if you vary any one of those parameters?

Mathematical Model

- $L(r)=L_W -20\log(r)-11-\alpha r$
- $r=\sqrt{x^2+h^2}$

Note: The term $-20\cdot\log_{10}(r)$ comes directly from the inverse-square law, which noise follows regarding distance from its source. Every time r doubles, this term increases by 6 dB, meaning the received level drops by 6 dB. At close range this term changes rapidly; at long range it flattens out, which is why the noise-vs-time curve has a sharp peak near overhead and long, gradual tails on either side.

Variable and Constant Definitions

Symbol	Name	Units
L(r)	Sound pressure level at the receiver	dB_0
L_W	Source sound power level	dB_f
r	straight-line distance from aircraft to receiver	m
x	Horizontal position of aircraft relative to receiver	m
h	Aircraft altitude (relative to reciever)	m
α	Atmospheric absorption coefficient	dB/m
$20\cdot\log_{10}(r)$	Geometric spreading loss accounts for energy diluting over a growing spherical wavefront	dB
11	Free-field correction constant for a point source radiating into open air	dB

Algorithm / System Logic

- **Step-by-step logic**
- At each timestep, compute the slant range r from the aircraft's current position to the ground receiver: $r = \sqrt{(x^2 + h^2)}$
- Plug r into the propagation equation to get the noise level:
    - $L(r) = L_W - 20 \cdot \log_{10}(r) - 11 - \alpha \cdot r$
- Store the current values of t, x, and L(r)
- Advance the aircraft: $x = x + v \cdot dt$, $t = t + dt$
- Repeat until $x > 5000$ m
    - There is one if/else statement after computing L(r) which clamps the model to zero if it comes out negative
    - Negative dB levels exist, but cannot be percieved by the human ear.

MATLAB Implementation

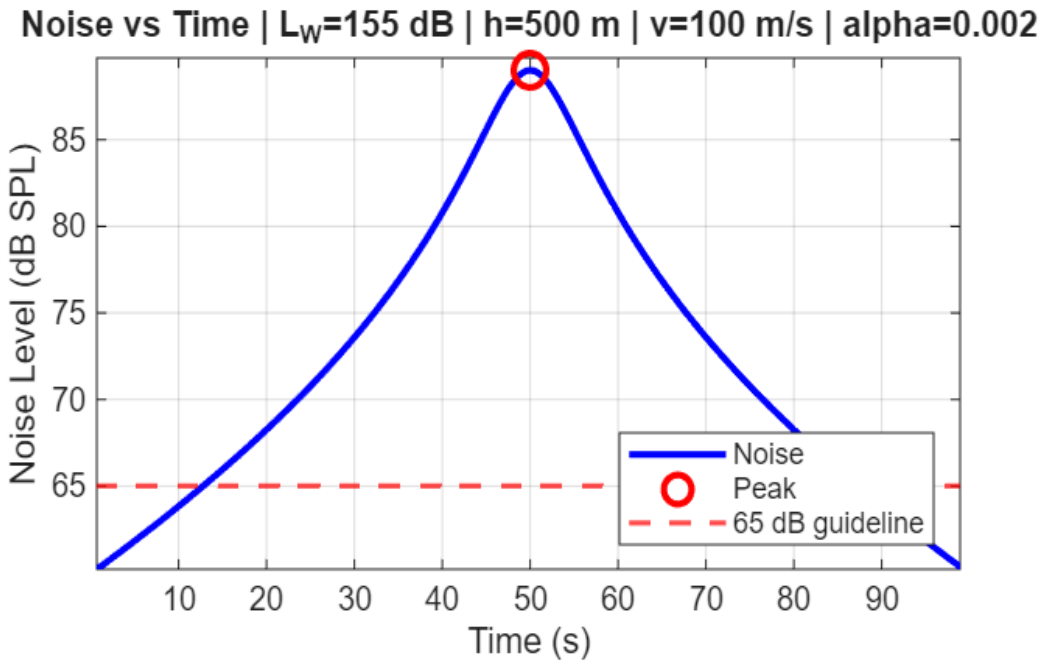
```
% aircraft altitude
fprintf('\nAltitude\n');
fprintf('  Landing approach  ~300 m\n');
fprintf('  Low overflight    ~600 m\n');
fprintf('  High overflight   ~3000 m\n');
h = input('  Enter altitude [m] (e.g. 500): ');
while ~isnumeric(h) || ~isscalar(h) || h <= 0 || h > 15000
    fprintf('  Invalid. Enter a positive value up to 15000 m.\n');
    h = input('  Enter altitude [m]: ');
end
```

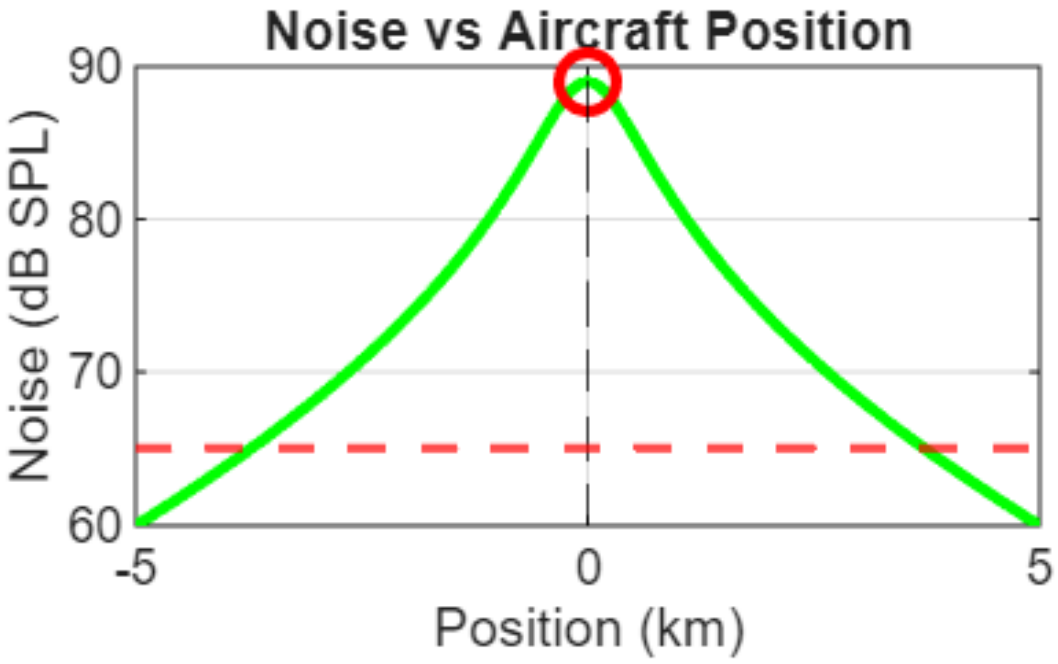
```
% Slant range from aircraft to receiver on ground
%   r = sqrt(x^2 + h^2)
r = sqrt(x^2 + h^2);

% Point-source propagation equation
%   L(r) = L_W - 20*log10(r) - 11 - alpha*r
dB = L_W - 20*log10(r) - 11 - alpha*r;
```

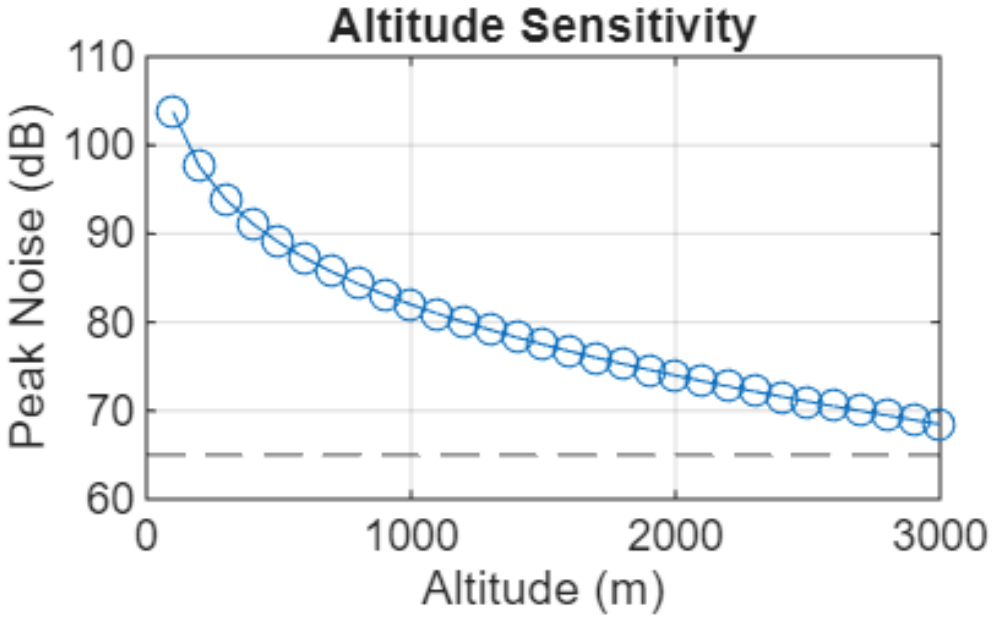
```
%% peak noise vs. altitude
%    (Fixed L_W and alpha; sweep altitude)

alt_range    = 100:100:3000;              % altitudes to test [m]
peak_vs_alt  = zeros(size(alt_range));

for k = 1:length(alt_range)
    h_k   = alt_range(k);
    x_k   = -5000;
    dB_max = 0;
    while x_k <= x_end
        r_k     = sqrt(x_k^2 + h_k^2);
        dB_k    = L_W - 20*log10(r_k) - 11 - alpha*r_k;
        dB_max  = max(dB_max, dB_k);
        x_k     = x_k + v*dt;
    end
    peak_vs_alt(k) = dB_max;
end
```

```
dt    = 0.5;          % timestep [s]
x_end = 5000;         % stop when aircraft reaches +5000 m [m]
x = -5000;            % initial horizontal position [m]
t = 0;                % initial time [s]
```

Simulation Results (Plots)

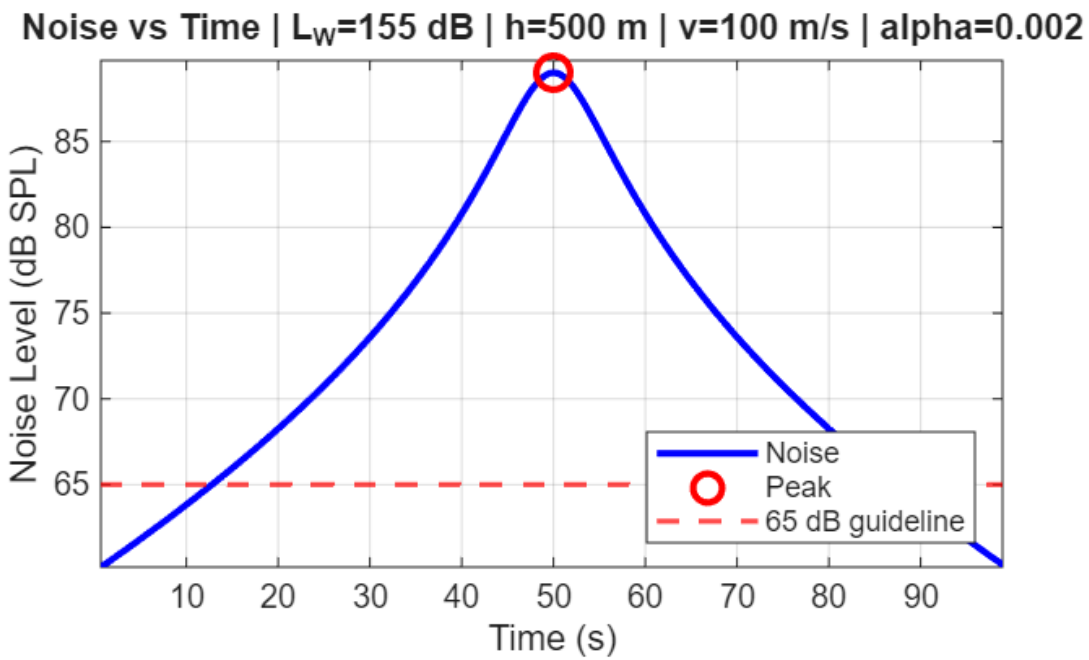






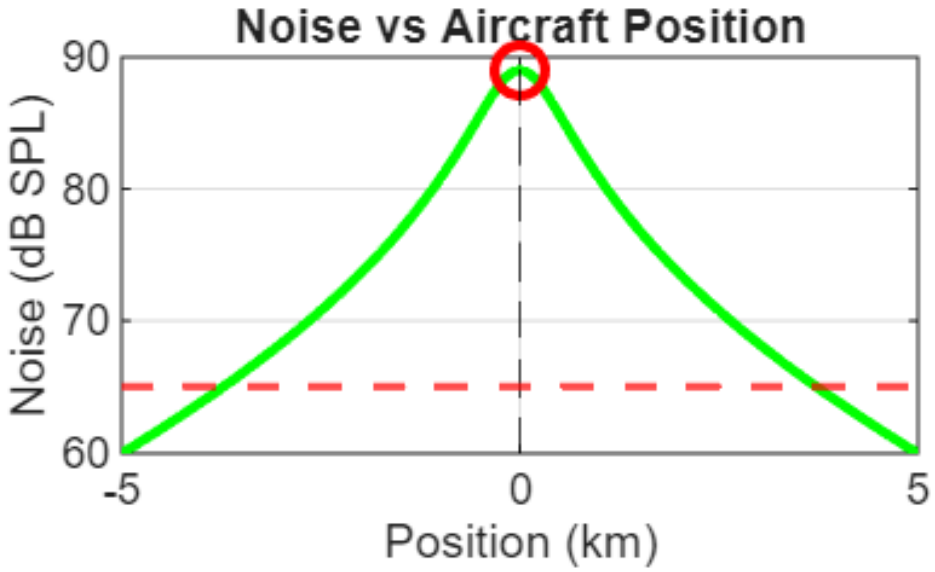
Analysis of Results

- **Noise vs Time**
  - The plot represents the noise vs time graph for the user-entered values
  - The graph represents the peak noise
  - Red line represents the 65 dB guideline



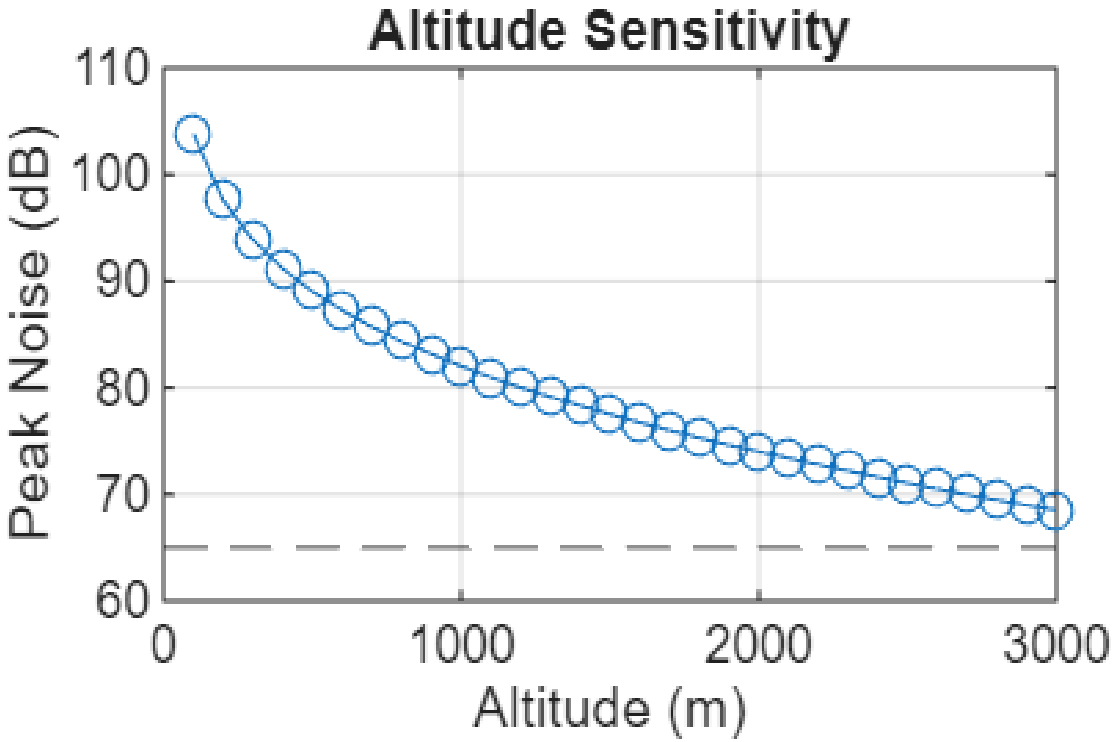
Analysis of Results

- **Noise vs Aircraft Position**
  - The closer to the aircraft, the higher the noise
  - The peak noise level is when the aircraft is 0 km away
  - Red line represents the 65 dB guideline



Parameter Sensitivity / Exploration

- **Altitude sensitivity**
  - Changing the altitude parameter
  - The peak noise decreases as altitude increases
  - The peak noise approaches the mandated 65dB limit (represented by dotted line)



Model Limitations

- **Assumptions**
  - Sound power output (L_W) is constant; thrust, throttle, and engine noise do not change
  - The atmosphere is uniform so that α is a single fixed number
  - The ground is flat and open for no terrain, buildings, or barriers between the aircraft and receiver
  - There is only one receiver so the model computes noise at a single ground point, not across an area.

- **Limitations**
  - Ground reflection: in reality, sound bounces off the ground and arrives at the receiver twice, which can add up to 6 dB depending on surface type
  - Directivity: jet engines are significantly louder to the rear than the front; the model assumes equal radiation in all directions
  - A real airport has dozens of simultaneous operations; this model handles only one aircraft at a time

Engineering Relevance

- **Why does this matter in real engineering?**
- This model could be used to observe the noise created by aircraft and come up with solutions, for either the aircraft itself or the runways, that can minimize the amount of noise around airports.
- Could be implemented for solving other noise issues that may be similar

Team Contributions

- What did each member do?
- Examples:
  - Member 1: Model Development
  - Member 2: MATLAB coding
  - Member 3: Analysis & visualization

Conclusion

- Summarize:
  - Built a system that models a noise monitor exposed to aircraft, which is moving along a straight horizontal path with constant speed and altitude
  - The model is used to find the peak noise level of the aircraft and then observe how that is impacted by altering the parameters.
  - Representative of the different ways matlab can be used for engineering